



AN OUTLIER IS **NOT** AUTOMATICALLY AN ERROR.

It could be bad data, a rare event, or your earliest warning signal.

Investigate. Don't ignore.

Outliers often tell the most important stories.

DATA ERROR



e.g., a value entered incorrectly (850 instead of 85).

RARE BUT REAL EVENT



e.g., a disease outbreak, a successful campaign, or a stock surge.

EARLY-WARNING SIGNAL



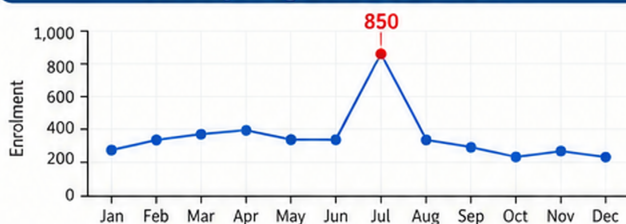
e.g., an unexpected increase in cases, drop in sales, or system failure.

ACTIONABLE INSIGHT



Verify, understand the cause, and use it to inform decisions and monitoring.

A REAL-WORLD EXAMPLE Monthly Programme Enrolment

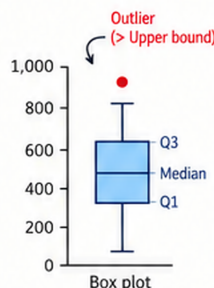


July shows 850 enrolments, much higher than other months.
Is this a data error? A successful campaign? Or the start of a new trend?

Don't remove it blindly. Investigate and understand the context.

HOW TO DETECT OUTLIERS USING THE IQR METHOD

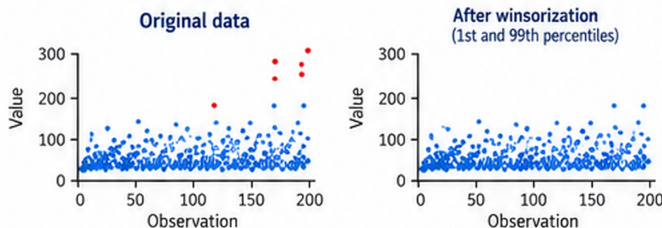
- 1 Calculate Q1 and Q3**
(Q1 = 25th percentile, Q3 = 75th percentile)
- 2 Compute the interquartile range (IQR)**
 $IQR = Q3 - Q1$
- 3 Define the outlier thresholds**
Lower bound = $Q1 - 1.5 \times IQR$
Upper bound = $Q3 + 1.5 \times IQR$
- 4 Flag values outside these bounds**
as potential outliers (not automatically errors).



- ✓ The IQR method is simple, robust, and widely used. It works well for many real-world datasets.

OPTIONAL TREATMENT: WINSORIZATION

Winsorization limits extreme values by capping them at a chosen percentile (e.g., 1st and 99th). It reduces the influence of outliers without removing data.



Winsorization is a tool, not a rule. Always understand the cause of outliers and consider whether to investigate, model separately, transform, or winsorize.

USE CASES BEYOND HEALTH



Public Health

Sudden increase in disease cases, unusual dropout rates, or unexpected vaccine demand.



Business & Sales

Unusual spike or drop in monthly sales, website traffic, or customer sign-ups.



Operations

Anomalies in system logs, equipment failure, or supply chain delays.



Finance

Fraud detection, unusual transactions, or market shocks.



Environment & Climate

Extreme weather events, unusual temperature or rainfall patterns.

THE EVIDENCELAB WORKFLOW FOR OUTLIERS



KEY TAKEAWAYS

- ✓ Outliers are not always errors — they can be valuable information.
- ✓ Use the IQR method to detect potential outliers.
- ✓ Winsorization can reduce the influence of extreme values.
- ✓ Always investigate and understand the context.
- ✓ Outliers and anomalies can provide early-warning signals.
- ✓ The same methods apply in health, business, finance and beyond.



COMMON MISTAKES

- Removing outliers without understanding why.
- Assuming all outliers are data errors.
- Using arbitrary cut-offs.
- Ignoring outliers in time-series monitoring.
- Failing to document your decisions.



TRY IT YOURSELF!

- ✓ Open the Google Colab notebook.
- ✓ Run all cells (no setup required).
- ✓ Explore real-world examples.
- ✓ Try IQR detection and winsorization.
- ✓ Apply anomaly detection to time-series data.



Open in Colab

Scan the QR code or click the link.

